

Diego Aguilar

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EDUCATION

Bachelor of Science in Computer Science

August 2022 – May 2026

University of Florida, Gainesville, FL

- GPA: 3.71 / 4.00
- Relevant Coursework: Applications of Discrete Structures, Data Structures & Algorithms, Introduction to Computer Organization, Introduction to Software Engineering, Object-Oriented Programming, Study Abroad UF In Japan: Cross Cultural Design, Operating Systems.

PROFESSIONAL EXPERIENCE

Teaching Assistant

January 2023 – Present

COP3503C | *University of Florida*, Gainesville, FL

- Managed a lab section which elaborated on core computer science topics, as well as demonstrated practical applications of concepts.
- Assisted students on projects such as Minesweeper, while working on the backend to develop an auto-grader for the assignment.
- Contributed on the backend of the course, helping with tasks ranging from course planning and assessments to creating labs.

Teaching Assistant

September 2024 – Present

Dr. Laura Cruz Castro | *University of Florida*, Gainesville, FL

- Worked directly with the professor of a python course and a graduate data science course to develop and test curriculum materials.
- Analyzed course content to develop lecture slides, homework assignments, and assist in the creation and revisions of exams.

PROJECTS

DGE (Diego Game Engine)

May 2021 – Present

- Developed good habits and learned best practices for improving code modularity in a big project.
- Learned many low level aspects of a game engine such as render pipelining by using only C++ and OpenCL.
- Integrated OOP build patterns, such as fluid builders, to streamline creation of many large scale backend systems within the project.

Minesweeper Autograder

September 2024

COP3503C - TA | *University of Florida*, Gainesville, FL

- Piloted an auto-grading library that when linked to student submissions, produces a score as well as a point breakdown.
- Leveraged design patterns to keep track of code state and strategically organized the large code base of the library.

Disaster Family

June 2024

CEN4930 | *University of Florida*, Kyoto, Japan

- Collaborated with a team to create a game focused on Cross Cultural Design with the theme of disaster prevention.
- Integrated core mechanics utilizing design patterns, including decorators, commands, and mediators.

Spell Checker

November 2023

COP3530 | *University of Florida*, Gainesville, FL

- Lead a team to create a spell checker used to compare two Data Structures in terms of performance, complexity, and efficiency.
- Developed a GUI to assist in the demonstration of the project, including a Trie traversal visualization in SFML.

Itch.io Games

May 2020 – Present

- Strengthened familiarity with Unity and C# by designing and programming various games and participating in game jams.
- Planned and coded the implementations of various features in original games utilizing Unity Scripting API.

SKILLS

Programming Languages: C, C++, C#, Java, Javascript, Python, SQL, HTML, CSS, MATLAB, AutoHotkey, LaTeX, ARM Assembly, OpenCL, Makefile, CMake, Rust.

Other Skills/Awards: Command Prompt, Unix, Design Patterns, Machine Learning, Unity, SDL2, SFML, Git, GitHub, GitHub Actions, Databases, Blender, Photoshop, Premiere, Jupyter Notebook, UF Dean's List.